Template Week 5 – Operating Systems

Student number:556707

Assignment 5.1: Unix-like

a) Find out what the difference is between UNIX and unix-like operating systems?

- UNIX**: Originally, UNIX refers to a family of operating systems that conform to the Single UNIX Specification and are typically certified by The Open Group. The original UNIX was developed in the 1970s at Bell Labs by Ken Thompson, Dennis Ritchie, and others. Certified UNIX systems strictly follow these standards.**

- Unix-like**: These are operating systems that behave similarly to UNIX and share many of its design principles but are not officially certified as UNIX. They mimic UNIX's interface and functionalities but may differ in internal design. Examples include Linux distributions, BSD variants (like FreeBSD), and macOS**.

b) the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources **often contain more detailed information about these individuals.**

- Ken Thompson**: Co-creator of UNIX at Bell Labs in the early 1970s. Developed the original UNIX kernel and the B programming language (predecessor to C).**

- Dennis Ritchie**: Collaborated with Thompson and invented the C programming language, which became the foundation for UNIX's portability and later many other systems.**

- Bill Joy**: A co-founder of Sun Microsystems, significant contributor to BSD UNIX development, creator of the vi editor, and early developer of the TCP/IP stack for BSD.**

- Richard Stallman**: Founder of the Free Software Movement and the GNU Project, which aimed to create a free UNIX-like operating system. He introduced the GNU General Public License (GPL), promoting software freedom.**

- Linus Torvalds**: Creator of the Linux kernel in 1991, which when combined with GNU software forms a fully functional free Unix-like operating system commonly called Linux.**

c) What is the philosophy of the GNU movement?

The GNU movement advocates for software freedom, emphasizing that users should have the freedom to:

- **Run the program for any purpose.**
- **Study and modify the source code.**
- **Redistribute copies.**
- **Distribute modified versions.**

The movement opposes proprietary software, promoting free (as in freedom) and open-source software to encourage collaboration, transparency, and user control over computing.

d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement? Please explain your answer.

Yes and no:

- **Yes: Ubuntu is a Linux distribution that includes GNU software and the Linux kernel. It is mostly free and open source, aligning with the GNU philosophy of software freedom.**
- **No: Ubuntu also includes some proprietary software/drivers and firmware to improve hardware compatibility and user experience, which slightly diverges from the strict GNU philosophy that rejects non-free software.**

e) Find out what is the Windows Subsystem for Linux?

WSL is a compatibility layer developed by Microsoft that allows users to run a genuine Linux environment directly on Windows without needing a virtual machine or dual boot. It enables developers to use Linux command-line tools, utilities, and applications on Windows systems seamlessly.

f) Find out, which operating system family belongs to Android, iOS and ChromeOS?

Android**: Based on the Linux kernel; belongs to the** Unix-like **family because it uses Linux as its kernel.**

iOS**: Based on Darwin (which is derived from BSD Unix); belongs to the** Unix **family.**

ChromeOS**: Also based on the Linux kernel; belongs to the** Unix-like **family.**

Assignment 5.2: Supercomputers and gameconsoles

a) Research on this site what supercomputers are used for and write a short summary of it:
https://www.computerhistory.org/timeline/search/?q=Supercomputer
Supercomputers are extremely powerful computers designed to perform complex calculations at very high speeds. They are used for tasks that require huge amounts of processing power, such as:
- **Scientific research and simulations (e.g., climate modeling, physics, astrophysics)**
- **National security applications (e.g., code-breaking, nuclear simulations)**
- **Engineering and industrial design (e.g., car crash simulations, aerospace design)**
- **Medicine and life sciences (e.g., drug development, molecular modeling)**
- **Artificial intelligence and big data processing**

Supercomputers enable scientists and engineers to solve problems that are too large or complex for regular computers by performing massive calculations quickly and efficiently.

b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for?
A PlayStation 3 cluster is a group (or "cluster") of multiple PlayStation 3 consoles linked together to work as a parallel computing system. Researchers used the PS3 consoles' powerful Cell processors—originally developed by IBM and Sony for high-performance computing tasks.
What was it used for?
- **The PS3 cluster was used for scientific research and high-performance computing experiments.**
- **Researchers leveraged the PS3's relatively affordable yet powerful hardware to create a cost-effective supercomputing cluster.**
- **Applications included simulating physics problems, molecular dynamics, protein folding, and other computationally intensive tasks.**
- **For example, in 2007, the US Air Force Research Laboratory built a PS3 cluster to perform weather simulations and other scientific calculations.**

Connection to IBM's Roadrunner:
- **The IBM Roadrunner supercomputer used a hybrid design with AMD Opteron CPUs and IBM's PowerXCell 8i processors; the latter evolved from the Cell processor developed for the PlayStation 3.**
- **The Cell processor's architecture was adapted and enhanced for high-performance computing in Roadrunner, bridging gaming hardware with supercomputing.**

c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example:
https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster
What specific operating system is running on this cluster?
The operating system running on the Oracle Raspberry Pi cluster is Oracle Linux (for ARM / "Oracle Linux for ARM").

d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list.
https://www.top500.org/lists/top500/list/2023/06/

Oracle's Raspberry Pi supercomputer does not **appear on the TOP500 list of the world's fastest supercomputers. Although it has many Raspberry Pi units, its overall computing power is far too low compared to the high-performance systems on the list, which require extremely fast processing measured in petaflops. The Raspberry Pi cluster is mainly for education and experimentation, not top-level supercomputing performance.**

e) What CPU architecture is used for the PlayStation 5 and Xbox Series X?
What operating systems run on these consoles?
What conclusion can you draw from the answer to the previous question?

The PlayStation 5 and Xbox Series X both use custom AMD CPUs based on the x86-64 (AMD Zen 2) architecture. This means their processors are similar to those found in modern PCs.
The operating systems running on these consoles differ:

- **The PlayStation 5 runs a custom operating system based on FreeBSD, a Unix-like system tailored for gaming performance.**
- **The Xbox Series X runs a custom OS built on the Windows 10 kernel, optimized for console use.**

Conclusion:
Both consoles use powerful PC-like CPU architectures, but their operating systems come from different families—PlayStation's is Unix-like, while Xbox's is Windows-based. This highlights how modern consoles balance hardware performance with software ecosystems suited to their respective platforms.

Assignment 5.3: Working with Windows

Take relevant screenshots of the assignments below

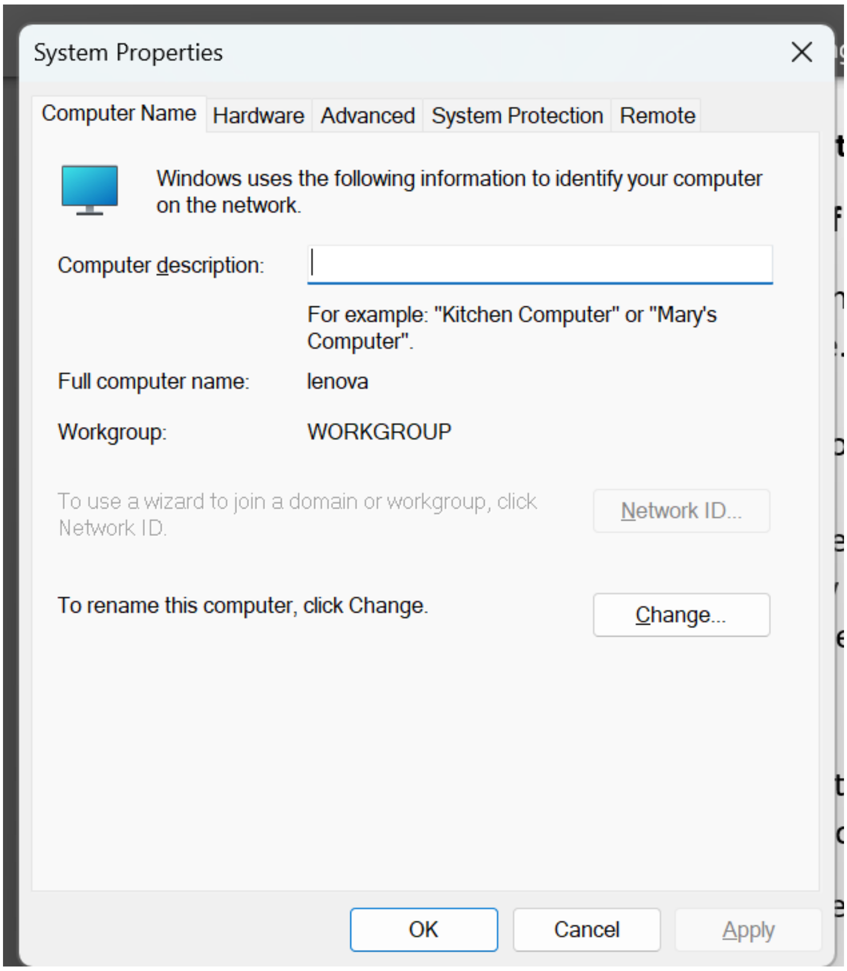
a) Practice for about 10 minutes with the keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.

b) The file explorer can be opened with + E, Which key combination could you also use?

- Windows + E will open File Explorer instantly
- Ctrl + N opens a new File Explorer window
- Ctrl + W will close the current File Explorer window

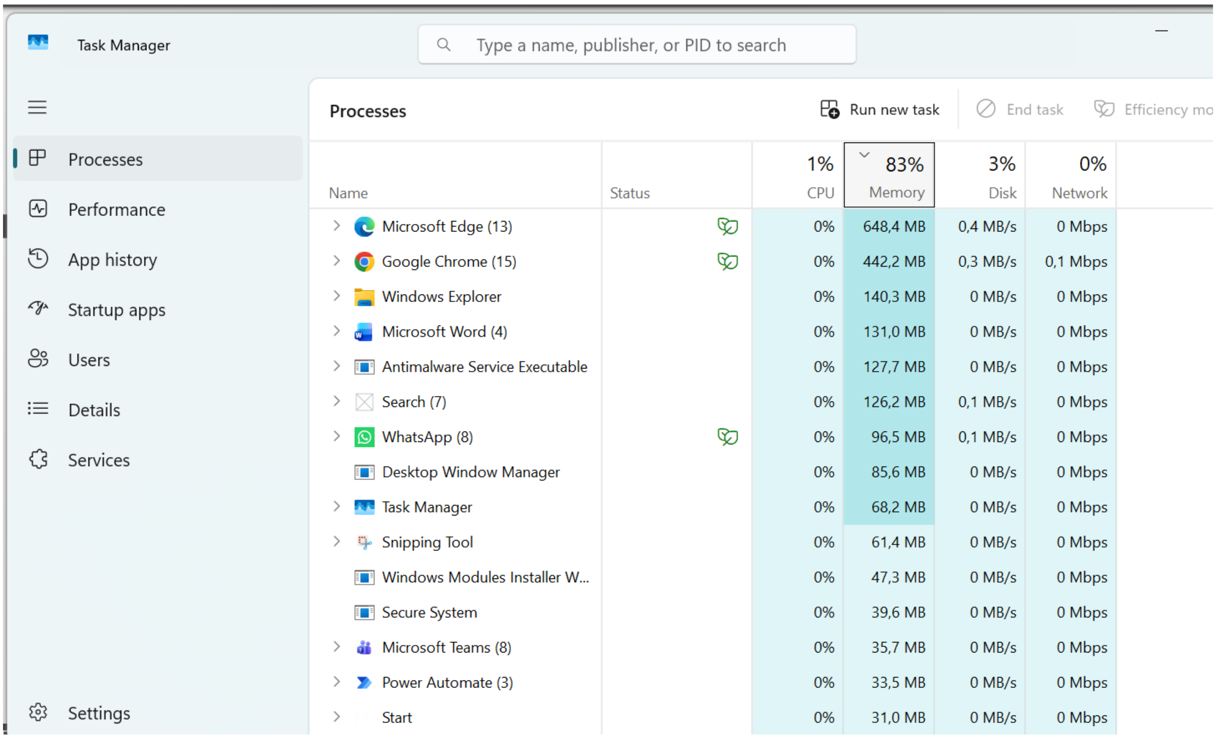
c) Open the system properties with a key combination, take a screenshot of the open screen. Paste this screenshot into this template.

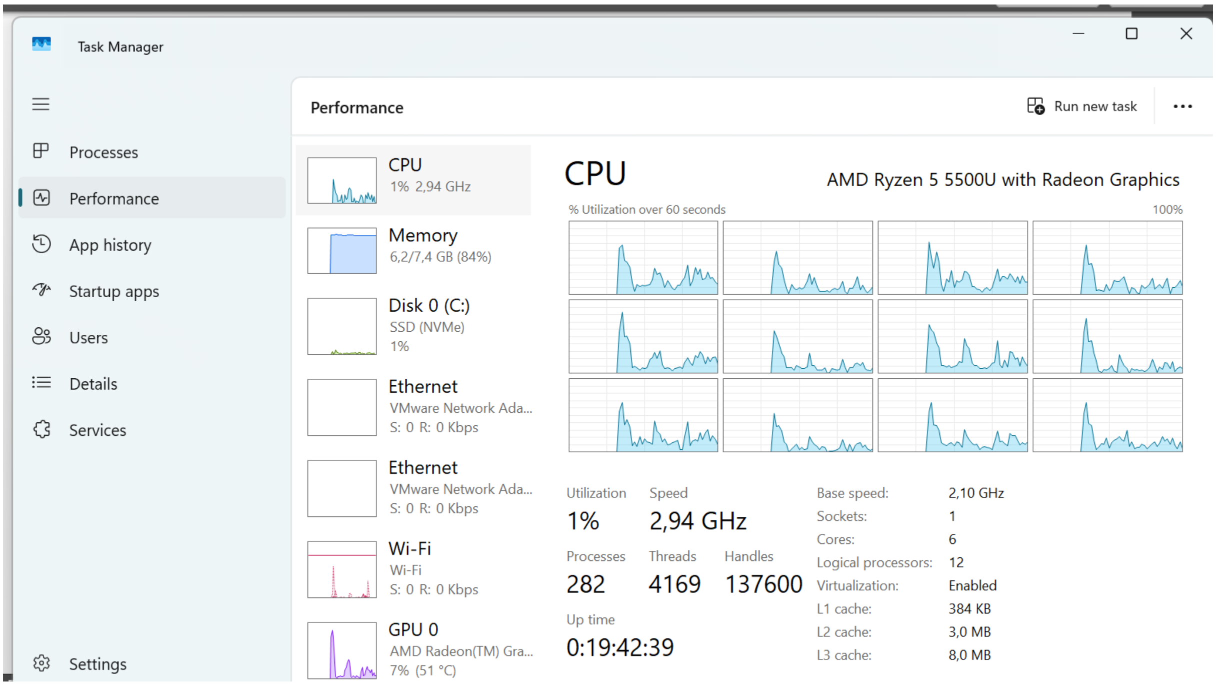
I can use Windows + R to open the "Run" and with sysdm.cpl I opens the System Properties dialog immediately.

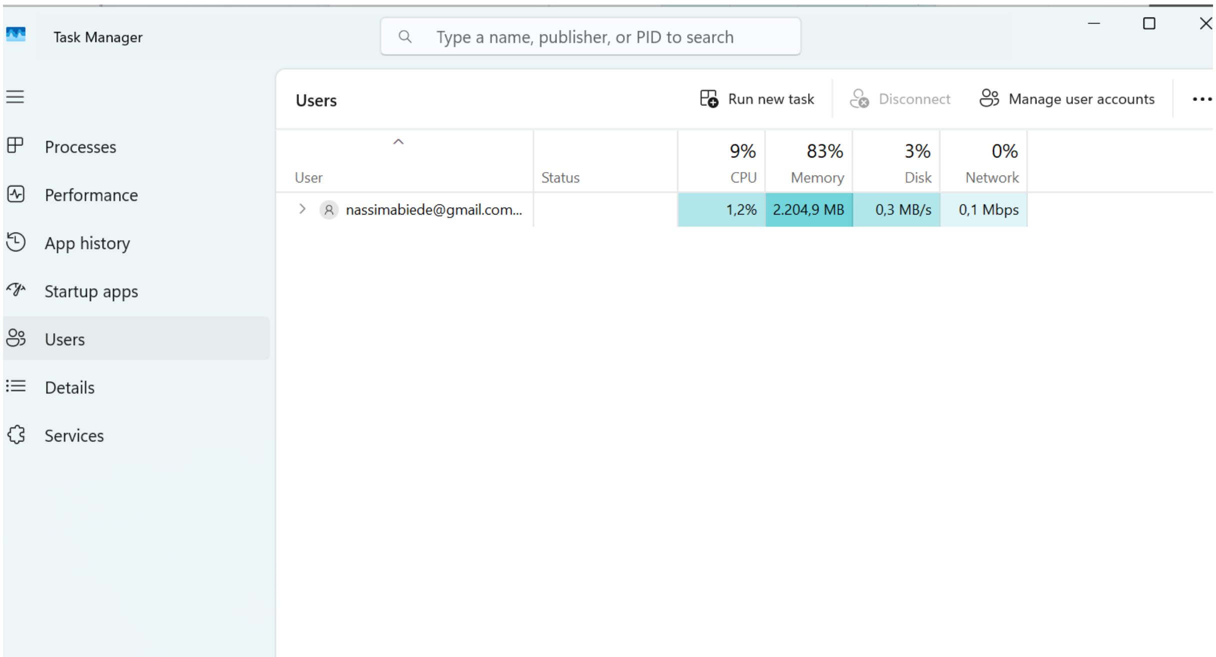


d) Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.

Shortcut for task manager is Ctrl + shift + Escape button.



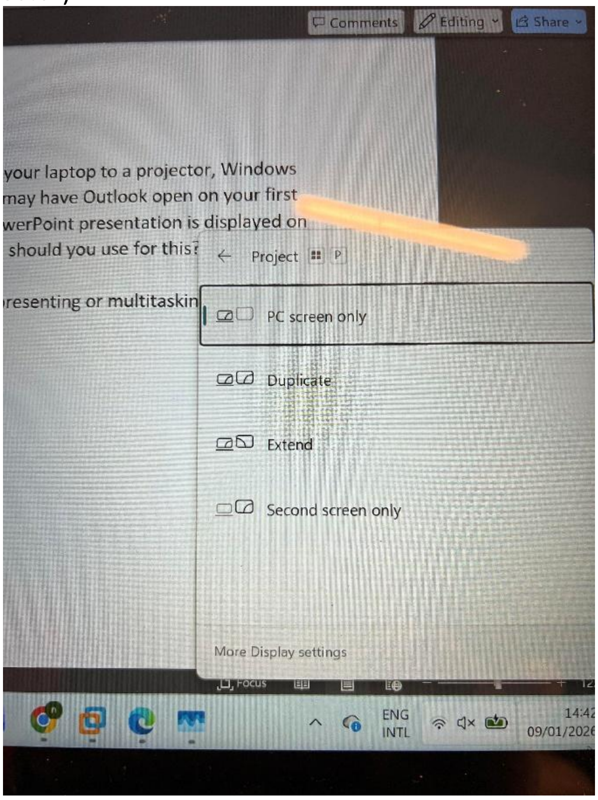




Here are the three screenshots of the Task manager.

e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?

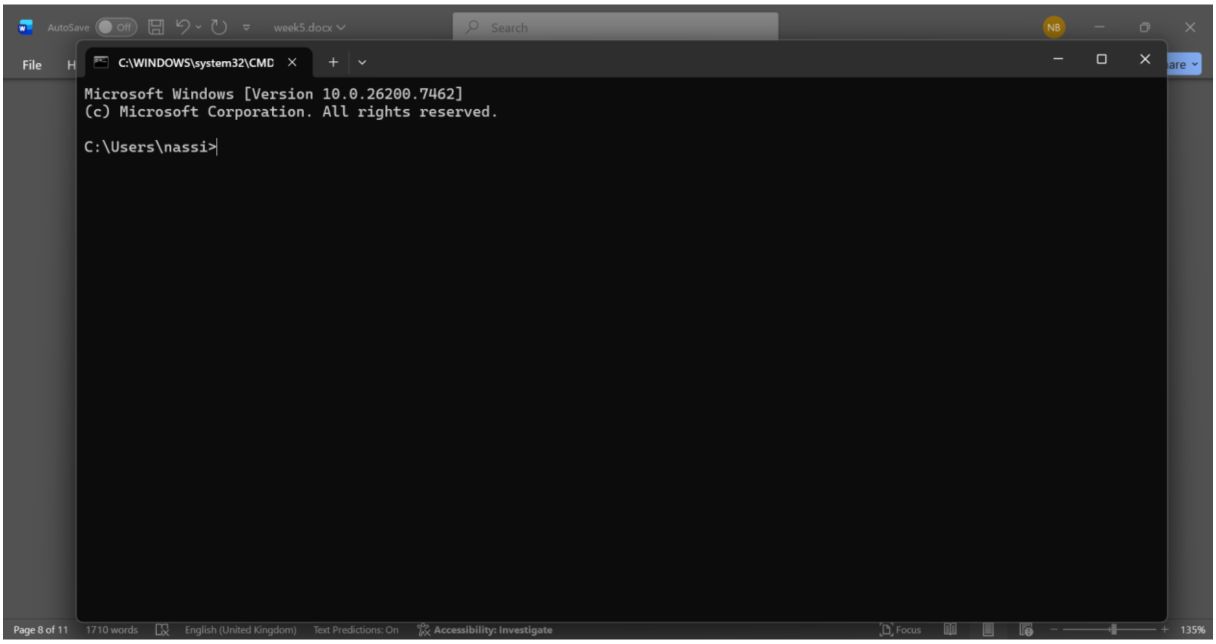
The shortcut to choose the right screen combination while presenting or multitasking is Start button and P (as projector).



f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?

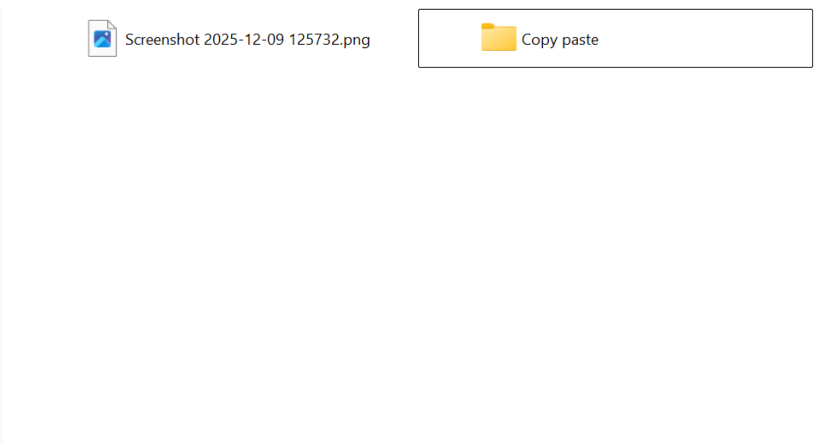
Start button + L will lock the screen.

g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.

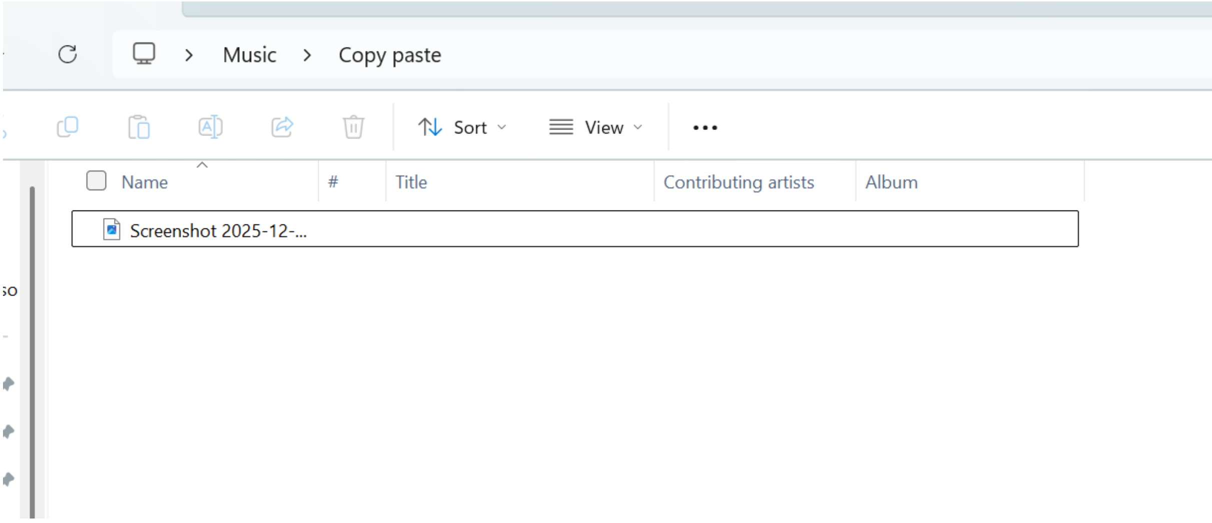


Working in the File Explorer

Relevant screenshots **copy** command:

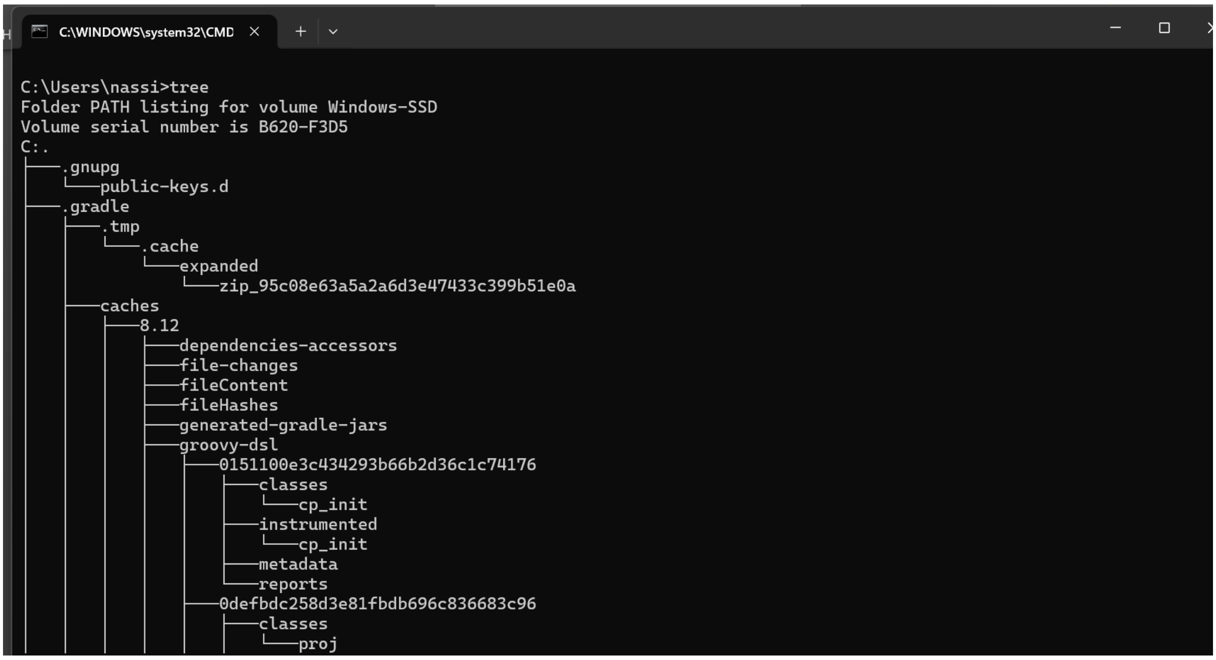


Here I have an screenshot, and by using Ctrl + C and Ctrl + V I move it to the "Copy paste" folder



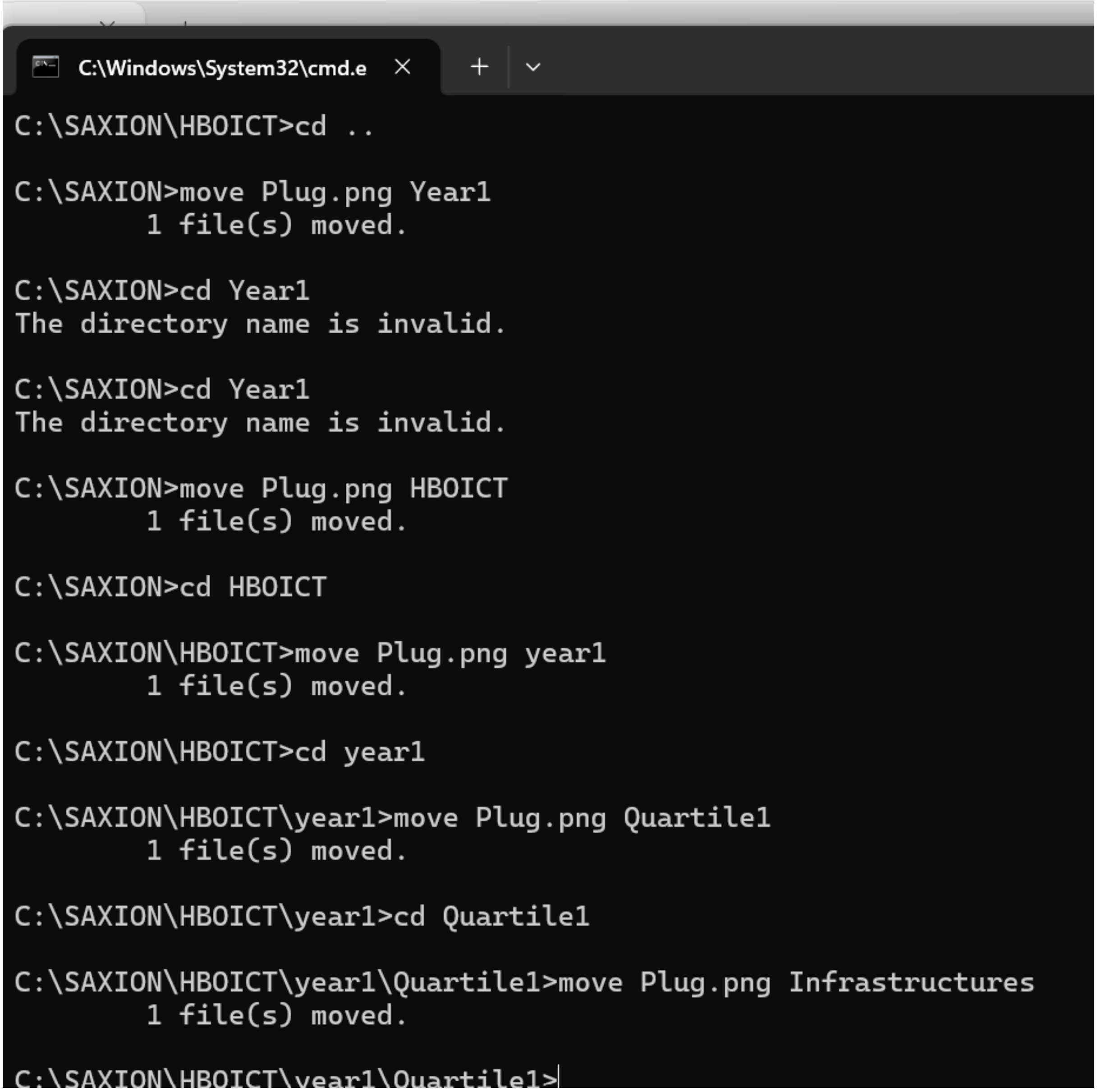
Relevant screenshots **tree** command:

The tree command in the Windows Command Prompt is used to display the directory structure of a s pecified path or the disk. It provides a clear hierarchical view of directories.



It currently shows all the files in C drive.

Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



This screenshots shows that I moved the 3 images using the move keyword In cmd.

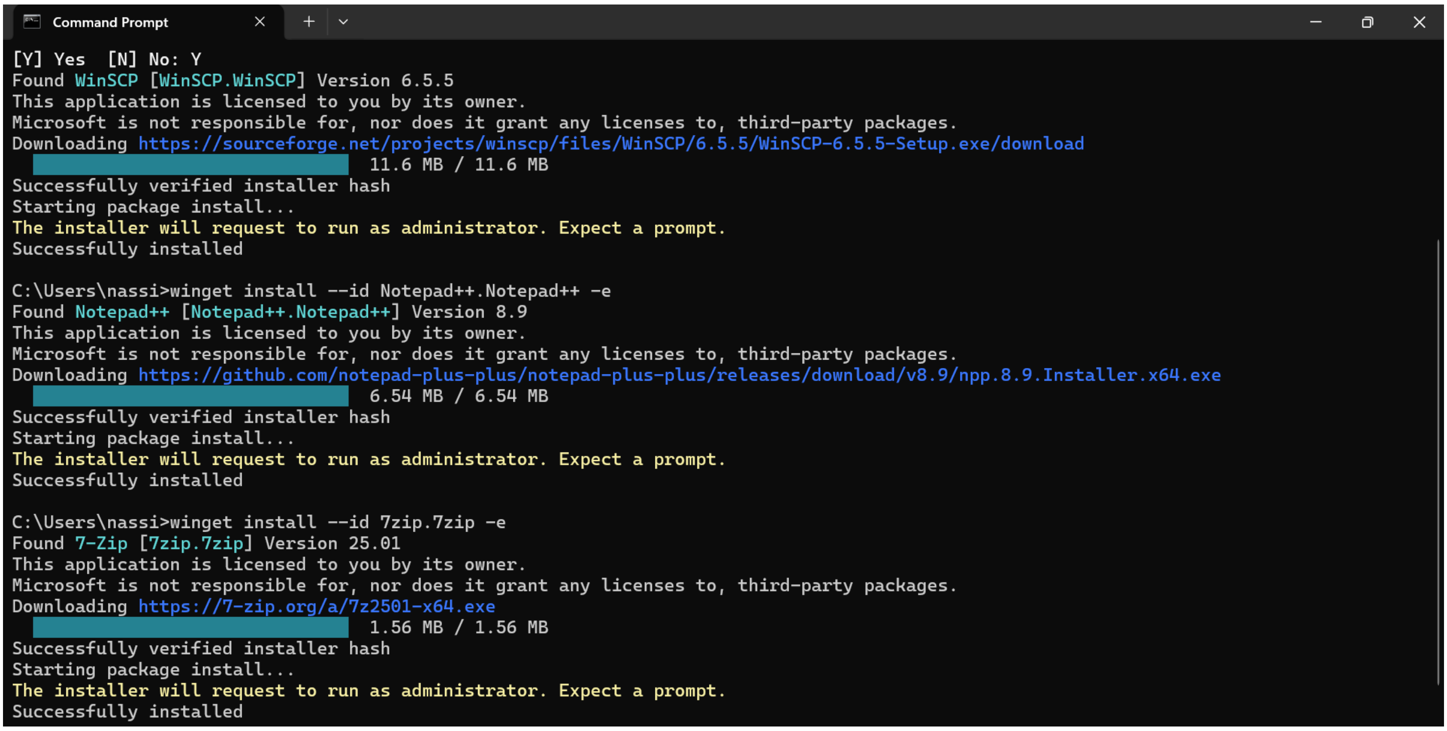
Terminating Processes

Relevant Screenshots Task Manager Window:

Install Software

Relevant screenshots that the following software is installed with winget:

- WinSCP
- Notepad++
- 7zip

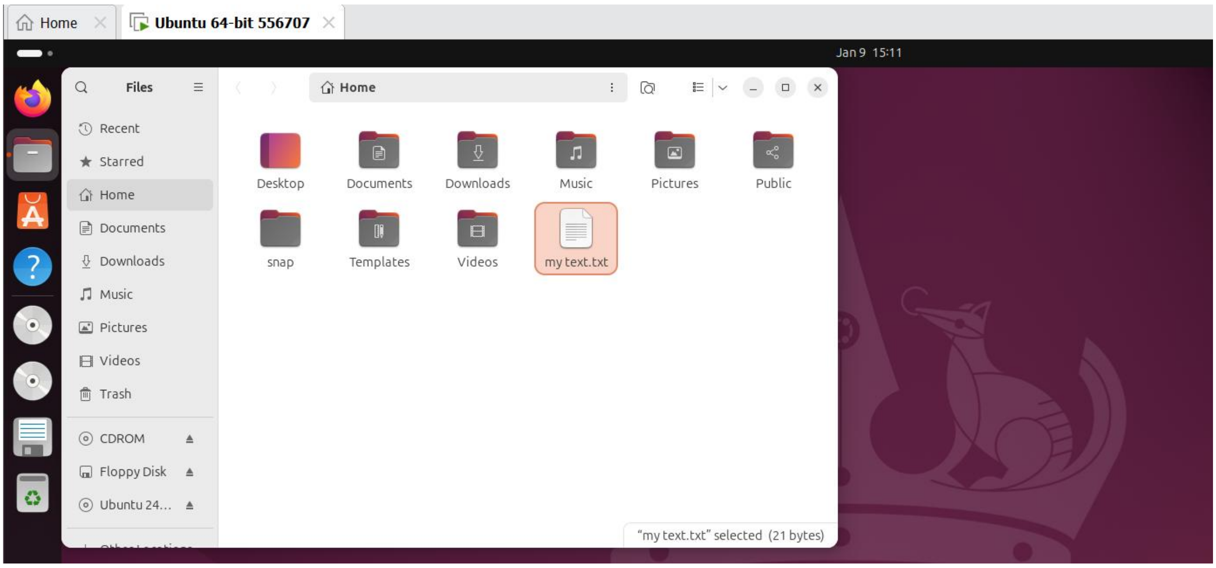


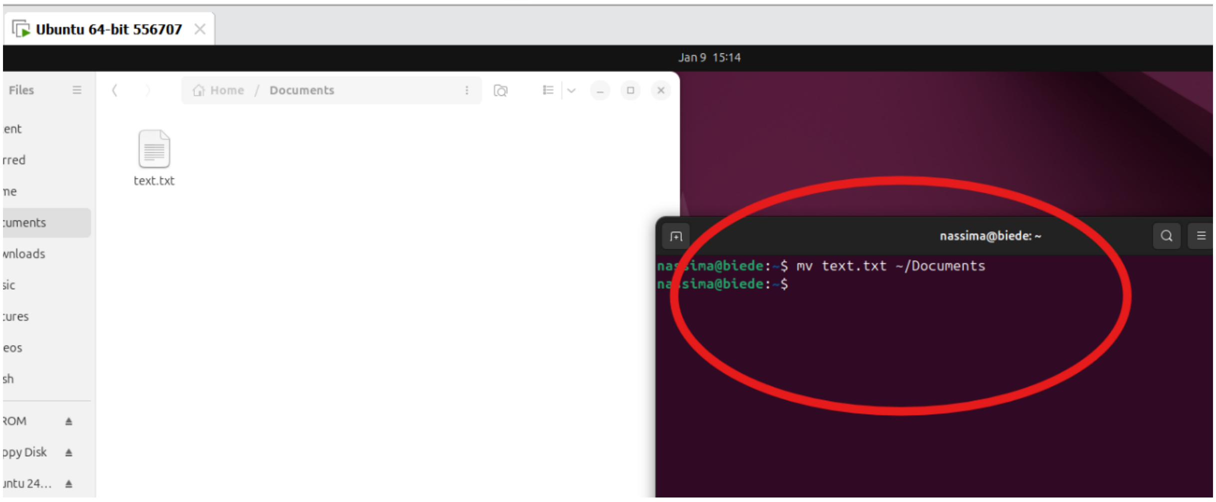
This screenshot is proof of installing the apps using the winget command in the cmd.

Assignment 5.4: Working with Linux

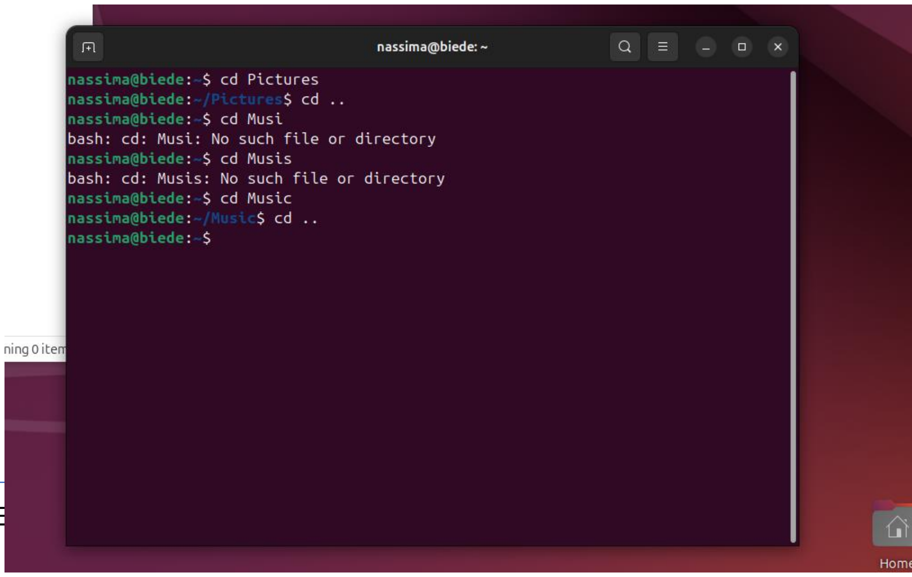
Relevant screenshots + motivation

1. Copying the text file using command line.





2. Navigating in linux



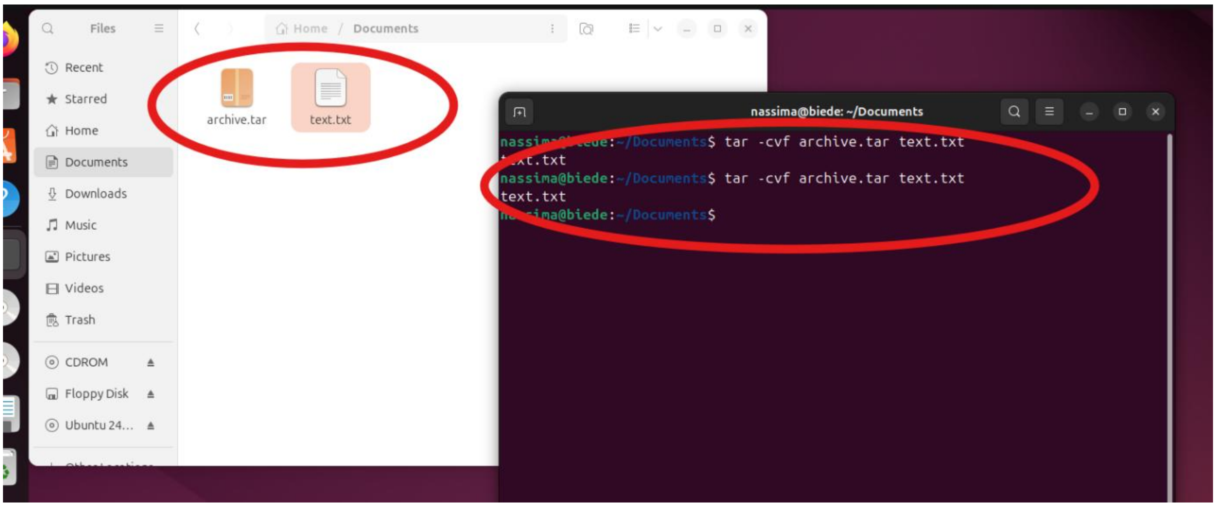
1.In a Linux terminal, you can return to your home directory by typing:
Cd or cd ~
Both commands take you to your home folder (e.g., /home/username).

2.Linux uses a single-root directory structure starting at /, whereas Windows uses multiple drive letters (such as C:, D:).

- Linux: everything is under /
- Windows: each drive has its own root (e.g., C:\)

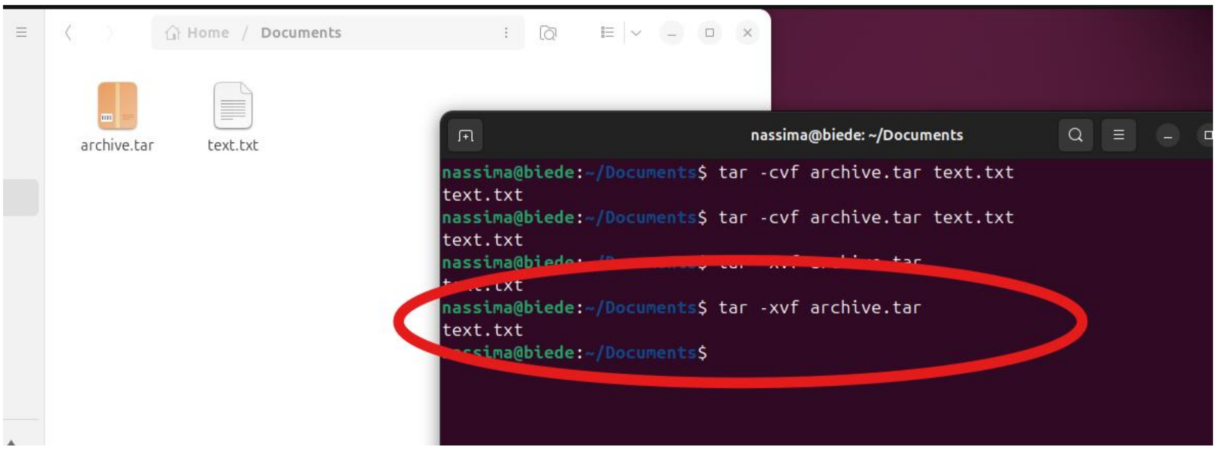
3.The /etc directory stores system-wide configuration files.
These files control how the system and its services behave (network settings, user accounts, startup services, etc.).

3. Compressing in Linux



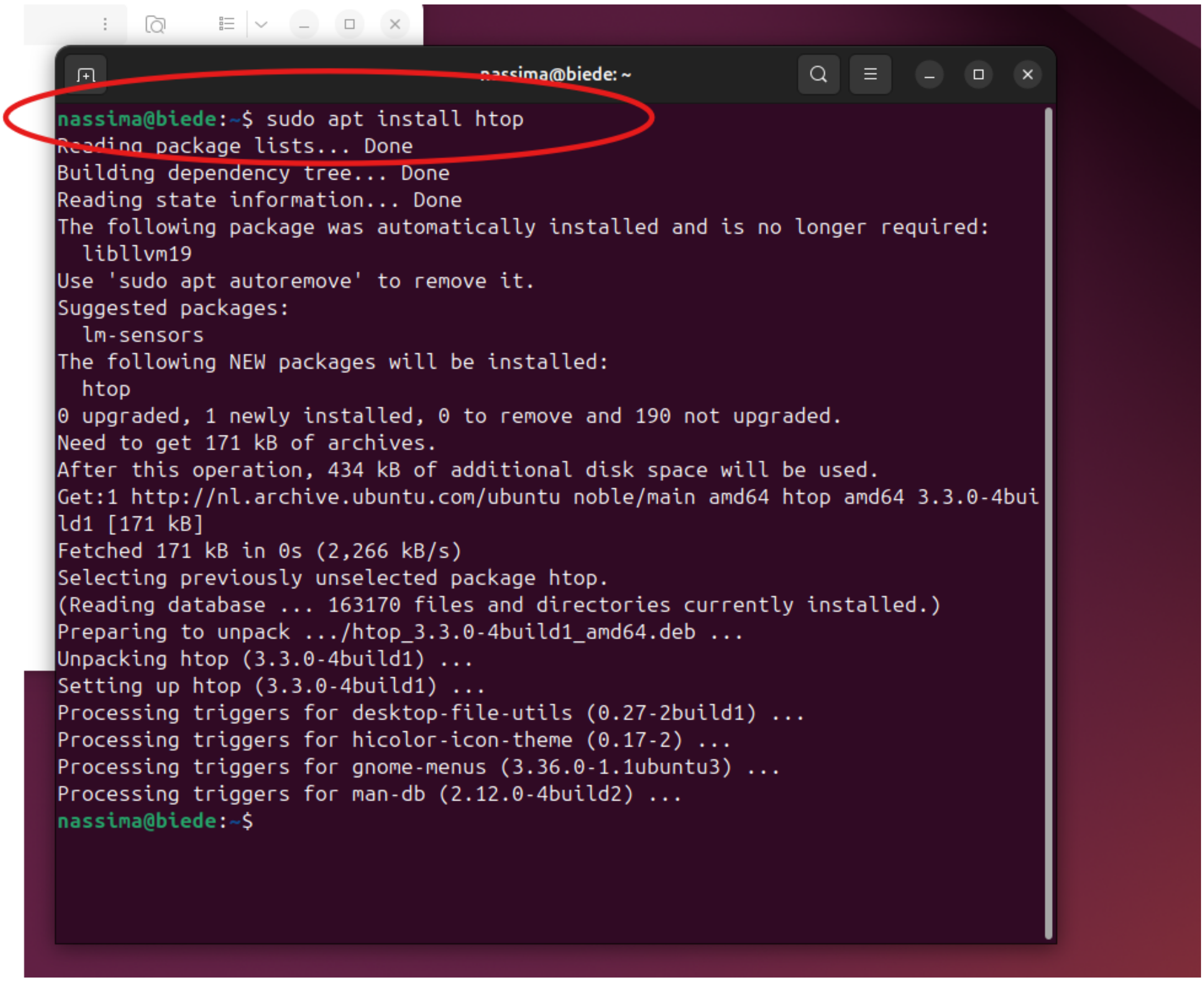
Command to create a tar archive from a text file:
tar -cvf archive.tar file.txt

- c = create archive
- v = verbose (shows progress)
- f = filename of the archive

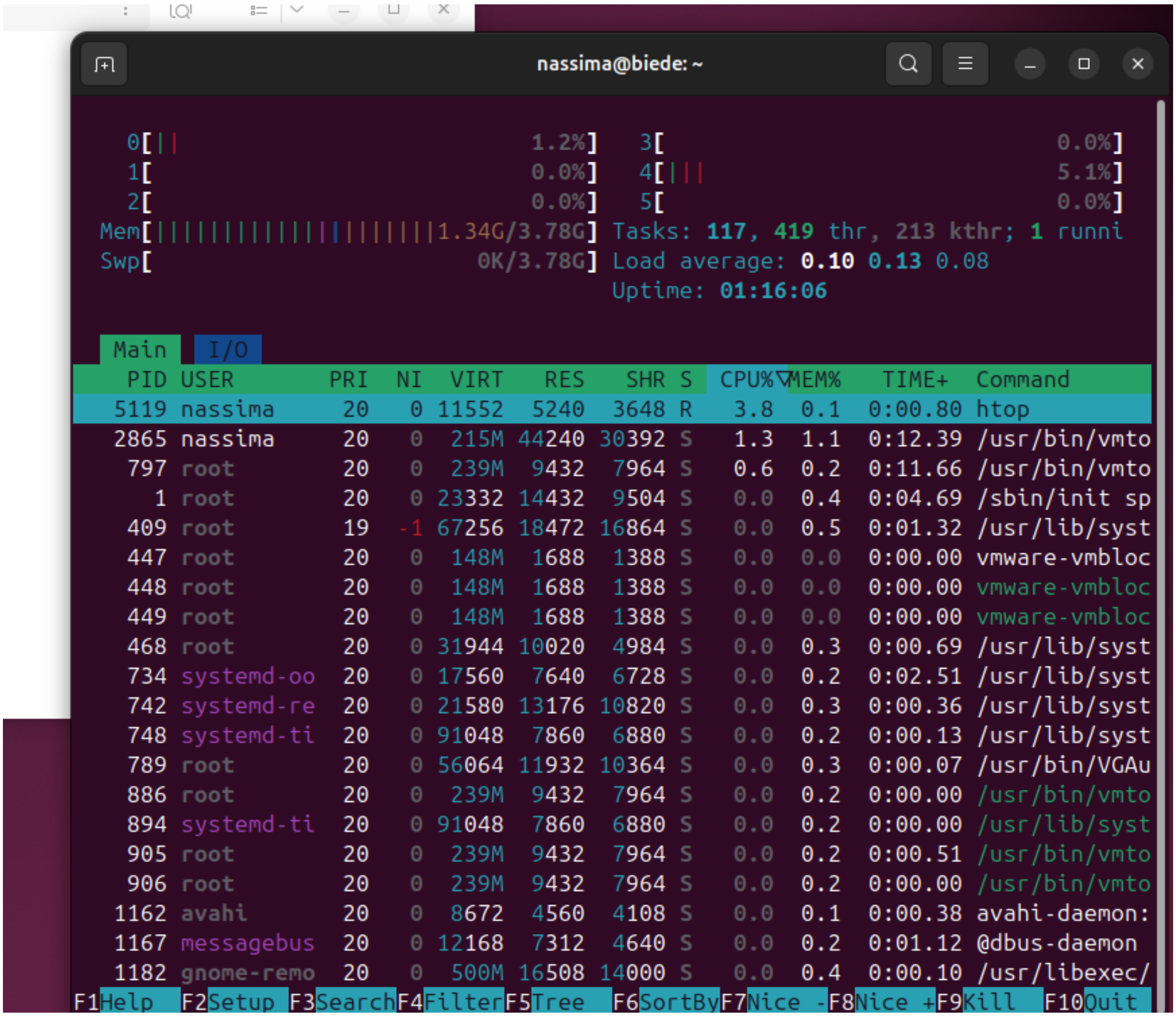


And now we extract the zip file that we created using *tar -xvf archive.tar* which Stands for **x = extract v = verbose f = archive file**

4. View proccesses



Installing the htop using apt install. This image below shows htop application that is used to monitor a real-time list of running processes, CPU usage, Memory (RAM) and swap usage Process IDs (PID), users, and priority, Ability to scroll, search, and kill processes.



Assignment 5.5: Users and permissions on Linux

Relevant screenshots + motivation

Assignment 5.6: View the contents of files

Relevant screenshots + motivation

Assignment 5.7: Digital forensics

Relevant screenshots + motivation

Assignment 5.8: Steganography

Relevant screenshots + motivation

Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.
- Proof that you can restore the back-up image into an empty VM.

Ready? Save this file and export it as a pdf file with the name: **week5.pdf**